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UniHack

AI-Powered Product Intelligence for
Industrial Commerce

Team Details

Team name: Team Statiz

Team leader name: Rahul G

Brief about your solution

Statiz turns minimal product information (part number, brand, short description, manufacturer URL, reference PDFs) into rich, governed product intelligence.

Given sparse inputs, the pipeline scrapes manufacturer pages and reference documents, extracts spec tables from PDFs, auto-classifies each product into UNSPSC taxonomy with a confidence score, generates the six description types, up to 20 features and 50 attribute triplets (label/value/UOM), and writes the exact 252-column delivery format — with a human review console for governance.

Result on demo data: 145 products, 100% family classification accuracy, 116 cross-supplier cross-references, 8-second CPU runtime, \$0 API cost.

1. How does your solution enrich minimal product information?

Briefly explain how your solution transforms limited inputs (e.g., Part Number, Brand, Short Description) into rich product intelligence.

2. How does your solution ensure accuracy and trust in the generated product data?

Describe your validation strategy. This may include:

(Confidence scoring, Multi-source verification, Human review, Rule-based validation, AI validation, Other approaches)

3. What makes your solution scalable for enterprise product catalogs?

Describe how your solution would handle: Large product catalogs, New manufacturers, Different document formats, Continuous product updates

Manufacturer page scraping (static + headless-browser fallback) extracts JSON-LD, meta tags and page text; reference PDFs (owner manuals, installation guides, spec sheets) are parsed into key/value tables. An LLM pass (structured JSON output) fuses all evidence into the delivery fields: 6 description variants, 20 features, 50 attribute triplets with UOMs, dimensions, certifications, images and document links. A heuristic JSON-LD path does the same offline.

2 · Accuracy and trust

- Confidence scoring on every classification (badge shown per product)
- Multi-source verification: LLM values never overwrite hard JSON-LD facts; attributes cross-checked between page and PDF
- Source grounding: every enriched attribute stores its source document + header
- Rule-based validation: unit/range checks, required-attribute flags per family
- Human review console: approve/correct before export — governed by construction
- Anti-hallucination prompt constraints: leave-blank rules; no invented certifications

3 · Enterprise scalability

- Fuzzy column detection (RapidFuzz): new manufacturers and unseen file layouts map automatically — no code changes, no hardcoding
- Embedding classification scales to full UNSPSC/eCI@ss (41k classes) by swapping taxonomy data — architecture unchanged

Opportunities

- a. How different is it from any of the other existing ideas?
- b. How will it be able to solve the problem statement given?
- c. USP of the proposed solution

How different from existing ideas?

No mainstream PIM/search vendor natively auto-classifies to industrial taxonomies (UNSPSC/eCI@ss/ETIM); only one (inRiver) partially extracts datasheets; enrichment leaders enrich their own search index, not an exportable governed catalog. Partly proved cross-referencing data is worth \$500M — but automotive-only. We combine all of it, self-serve.

How it solves the problem statement

Minimal product information (part no, brand, short desc) + manufacturer URLs is exactly our input; the 252-column delivery format is exactly our output — populated dynamically with unseen evaluation data handled by fuzzy schema detection.

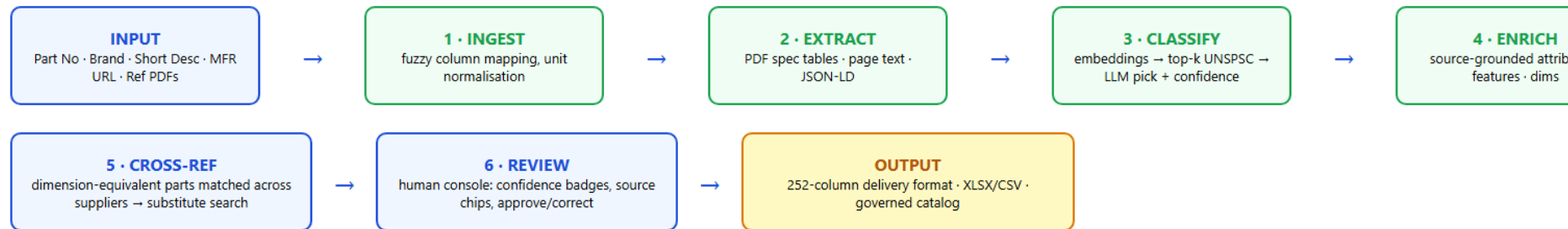
USP

1. \$0 cost, CPU-only, offline-capable (graceful LLM fallback) — vs \$45–100K/yr tools
2. Every AI value carries evidence: confidence + source span
3. Cross-supplier substitute matching built in
4. 8-second end-to-end runtime on a laptop

List of features offered by the solution

- Fuzzy schema ingestion: any supplier's CSV/XLSX column names auto-mapped (in→mm, comma decimals)
- Manufacturer URL scraping with headless-browser fallback · PDF spec-sheet extraction
- UNSPSC auto-classification with confidence scores and top-5 candidate evidence
- LLM enrichment: 6 description types, 20 features, 50 attribute triplets (label/value/UOM)
- Cross-supplier part cross-referencing + substitute search API
- Human review console: approve/correct, confidence badges, source-tagged attributes
- Delivery export: all 252 static headers populated exactly (CSV + XLSX)
- Multi-provider free LLM layer (Cerebras/Groq/Gemini) with deterministic offline fallback

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)

Statiz

Industrial product intelligence — review console

[Export CSV](#)
[Re-run pipeline](#)

145 PRODUCTS

10 SPEC PDFS

5 PDF-ENRICHED ATTRS

116 CROSS-REFERENCED

local embeddings

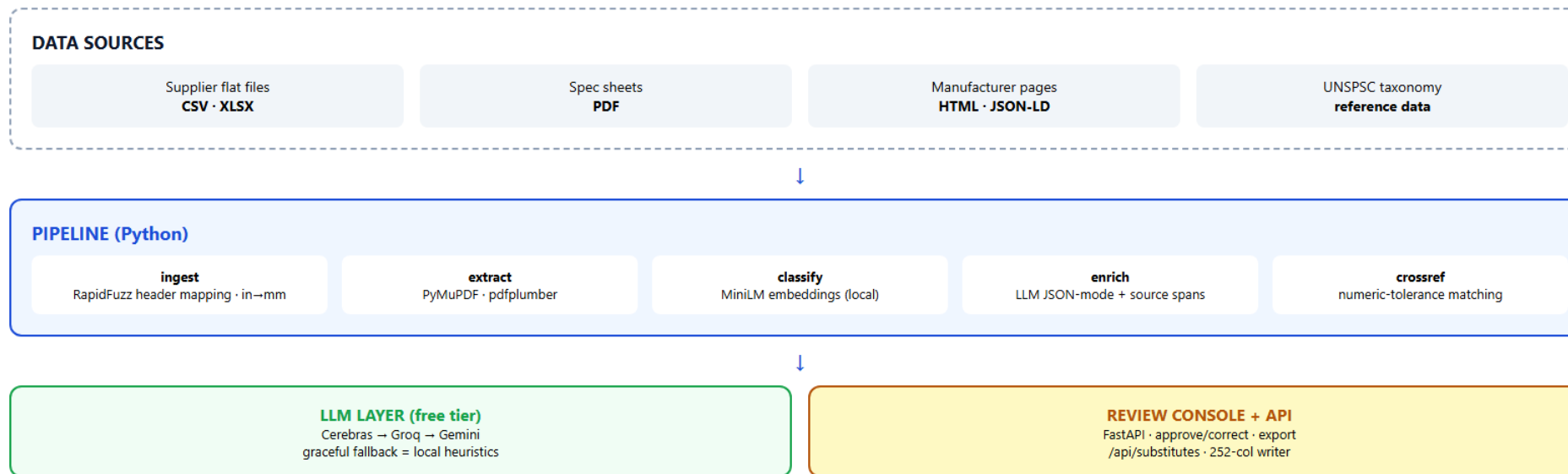
8.2 PIPELINE SEC

All families

Any confidence

PART	DESCRIPTION	UNSPSC CLASSIFICATION	ATTRIBUTES	PRICE	CROSS-REFS	REVIEW
BRG-BEA0000 supplierA_bearings.csv	Deep groove ball bearing 6001-2RS	Ball bearings 74% undefined - embedding	bore_mm: 25mm max_rpm: 9000 rpm od_mm: 45mm dynamic_load_kn: 14.3C seal_type: contact seal	45.24 €	7 matches	Approved
BRG-BEA0001 supplierA_bearings.csv	Deep groove ball bearing 6308-2RS	Ball bearings 74% undefined - embedding	bore_mm: 63mm max_rpm: 9000 rpm od_mm: 77mm dynamic_load_kn: 7.4C seal_type: contact seal	17.03 €	11 matches	Approve
BRG-BEA0002 supplierA_bearings.csv	Deep groove ball bearing 6308-2RS	Ball bearings 74% undefined - embedding	bore_mm: 63mm max_rpm: 15000 rpm od_mm: 77mm dynamic_load_kn: 12.8C seal_type: contact seal	49.33 €	11 matches	Approve
BRG-BEA0003 supplierA_bearings.csv	Deep groove ball bearing 6002-2RS	Ball bearings 73% undefined - embedding	bore_mm: 52mm max_rpm: 9000 rpm od_mm: 67mm dynamic_load_kn: 14.4C seal_type: contact seal	9.72 €	3 matches	Approve
BRG-BEA0004 supplierA_bearings.csv	Deep groove ball bearing 6208-Z	Ball bearings 74% undefined - embedding	bore_mm: 63mm max_rpm: 15000 rpm od_mm: 77mm dynamic_load_kn: 29.5C seal_type: shield seal	34.03 €	11 matches	Approve
BRG-BEA0005 supplierA_bearings.csv	Deep groove ball bearing 6305-2RS	Ball bearings 74% undefined - embedding	bore_mm: 62mm max_rpm: 15000 rpm od_mm: 77mm dynamic_load_kn: 30.1C seal_type: contact seal	36.33 €	11 matches	Approve
BRG-BEA0006 supplierA_bearings.csv	Deep groove ball bearing 6205	Ball bearings 73% undefined - embedding	bore_mm: 30mm max_rpm: 9000 rpm od_mm: 45mm dynamic_load_kn: 34.6C seal_type: open seal	25.29 €	11 matches	Approve
BRG-BEA0007 supplierA_bearings.csv	Deep groove ball bearing 6002-Z	Ball bearings 72% undefined - embedding	bore_mm: 30mm max_rpm: 15000 rpm od_mm: 45mm dynamic_load_kn: 16.6C seal_type: shield seal	43.30 €	11 matches	Approve
BRG-BEA0008 supplierA_bearings.csv	Deep groove ball bearing 6205	Ball bearings 73% undefined - embedding	bore_mm: 30mm max_rpm: 9000 rpm od_mm: 50mm dynamic_load_kn: 27.3C seal_type: open seal	25.25 €	5 matches	Approve
BRG-BEA0009 supplierA_bearings.csv	Deep groove ball bearing 6001	Ball bearings 72% undefined - embedding	bore_mm: 62mm max_rpm: 9000 rpm od_mm: 72mm dynamic_load_kn: 26.0C seal_type: open seal	50.02 €	7 matches	Approve
BRG-BEA0010 supplierA_bearings.csv	Deep groove ball bearing 6305-Z	Ball bearings 71% undefined - embedding	bore_mm: 25mm max_rpm: 15000 rpm od_mm: 35mm dynamic_load_kn: 32.5C seal_type: shield seal	15.91 €	1 match	Approve
BRG-BEA0011 supplierA_bearings.csv	Deep groove ball bearing 6001-Z	Ball bearings 72% undefined - embedding	bore_mm: 30mm max_rpm: 15000 rpm od_mm: 45mm dynamic_load_kn: 10.5C seal_type: shield seal	45.83 €	11 matches	Approve

Architecture diagram of the proposed solution



Technologies used in the solution

Python 3.14 · FastAPI + Uvicorn (API & review console) · sentence-transformers MiniLM (local embeddings)

PyMuPDF + pdfplumber (PDF extraction) · RapidFuzz (fuzzy matching) · pandas/openpyxl (I/O)

Playwright/Chromium (JS-page scraping) · httpx (cached fetch layer)

LLM layer: Cerebras → Groq → Gemini free tiers, OpenAI-compatible JSON mode

Estimated implementation cost (optional)

Prototype: \$0.

- LLMs: free tiers (Cerebras ~30K TPM / Groq / Gemini), offline heuristic fallback
- Embeddings: local on CPU — no API cost
- Hosting: Cloudflare quick tunnel (demo); Render free tier for a persistent MVP

At scale: paid LLM tier $\approx$ \$0.001–0.005 per product; 100K products $\approx$ \$100–500 one-time.

Snapshots of the MVP

Statiz Industrial product intelligence - view console						
145 PRODUCTS	10 SPEC PDOS	5 PDF-ENRICHED ATTRS	116 CROSS-REFERENCED	local embeddings LLM PROVIDERS	8.2 PIPELINE SEC	Export CSV Run pipeline
Search part number / description... All families Any confidence						
PART	DESCRIPTION	UNISPEC CLASSIFICATION	ATTRIBUTES		PRICE	CROSS-REFS
BRS BEA000 subpart_bearing.cdr	Deep groove ball bearing 6001-2RS	Ball bearings 74% unclassified embedding	bore_mm:25.000 mm id:rm:4500 mm dynamic_load_kn:1430 max_spm:1800 rpm shaft_dia_mm:25.000 contact seal:yes		45.24 €	7 matches Approved
BRS BEA001 subpart_bearing.cdr	Deep groove ball bearing 6300-2RS	Ball bearings 74% unclassified embedding	bore_mm:62.000 mm id:rm:7700 mm dynamic_load_kn:1540 max_spm:1800 rpm shaft_dia_mm:62.000 contact seal:yes		17.03 €	11 matches Approved
BRS BEA002 subpart_bearing.cdr	Deep groove ball bearing 6300-2RS	Ball bearings 74% unclassified embedding	bore_mm:62.000 mm id:rm:7700 mm dynamic_load_kn:1520 max_spm:1900 rpm shaft_dia_mm:62.000 contact seal:yes		49.33 €	11 matches Approved
BRS BEA003 subpart_bearing.cdr	Deep groove ball bearing 6002-2RS	Ball bearings 73% unclassified embedding	bore_mm:32.000 mm id:rm:6700 mm dynamic_load_kn:1440 max_spm:1800 rpm shaft_dia_mm:32.000 contact seal:yes		9.72 €	1 matches Approved
BRS BEA004 subpart_bearing.cdr	Deep groove ball bearing 6300-2	Ball bearings 73% unclassified embedding	bore_mm:62.000 mm id:rm:7700 mm dynamic_load_kn:2830 max_spm:1900 rpm shaft_dia_mm:62.000		34.03 €	11 matches Approved
BRS BEA005 subpart_bearing.cdr	Deep groove ball bearing 6300-2RS	Ball bearings 73% unclassified embedding	bore_mm:62.000 mm id:rm:7700 mm dynamic_load_kn:3030 max_spm:1900 rpm shaft_dia_mm:62.000 contact seal:yes		56.33 €	11 matches Approved
BRS BEA006 subpart_bearing.cdr	Deep groove ball bearing 6305	Ball bearings 73% unclassified embedding	bore_mm:32.000 mm id:rm:4500 mm dynamic_load_kn:3480 max_spm:1800 rpm shaft_dia_mm:32.000		25.29 €	11 matches Approved
BRS BEA007 subpart_bearing.cdr	Deep groove ball bearing 6002-Z	Ball bearings 72% unclassified embedding	bore_mm:32.000 mm id:rm:4500 mm dynamic_load_kn:1680 max_spm:1900 rpm shaft_dia_mm:32.000		41.30 €	11 matches Approved
BRS BEA008 subpart_bearing.cdr	Deep groove ball bearing 6305	Ball bearings 70% unclassified embedding	bore_mm:32.000 mm id:rm:5500 mm dynamic_load_kn:2730 max_spm:1800 rpm shaft_dia_mm:32.000		25.25 €	1 matches Approved
BRS BEA009 subpart_bearing.cdr	Deep groove ball bearing 6001	Ball bearings 70% unclassified embedding	bore_mm:62.000 mm id:rm:7700 mm dynamic_load_kn:2630 max_spm:1800 rpm shaft_dia_mm:62.000		50.02 €	7 matches Approved
BRS BEA010 subpart_bearing.cdr	Deep groove ball bearing 6300-Z	Ball bearings 70% unclassified embedding	bore_mm:25.000 mm id:rm:3500 mm dynamic_load_kn:3230 max_spm:1900 rpm shaft_dia_mm:25.000		15.91 €	1 matches Approved
BRS BEA011 subpart_bearing.cdr	Deep groove ball bearing 6001-Z	Ball bearings 70% unclassified embedding	bore_mm:32.000 mm id:rm:4500 mm dynamic_load_kn:1530 max_spm:1900 rpm shaft_dia_mm:32.000		45.43 €	11 matches Approved

```

Pretty-print
{"query": "BRS-BEA000", "substitutes": [{"part_number": "BRS-BEA001", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 56.46, "currency": "EUR"}}, {"part_number": "BRS-BEA002", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 19.6, "currency": "EUR"}}, {"part_number": "BRS-BEA003", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 49.33, "currency": "EUR"}}, {"part_number": "BRS-BEA004", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 34.03, "currency": "EUR"}}, {"part_number": "BRS-BEA005", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 56.33, "currency": "EUR"}}, {"part_number": "BRS-BEA006", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 25.29, "currency": "EUR"}}, {"part_number": "BRS-BEA007", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 41.3, "currency": "EUR"}}, {"part_number": "BRS-BEA008", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 25.25, "currency": "EUR"}}, {"part_number": "BRS-BEA009", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 50.02, "currency": "EUR"}}, {"part_number": "BRS-BEA010", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 15.91, "currency": "EUR"}}, {"part_number": "BRS-BEA011", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 45.43, "currency": "EUR"}}]}

```

Review console (left) · substitute search API (right)

Additional Details/Future Development (if any)

- Engineering-constraint substitution validation (fit/load margins, not just dimensions)
- OCR path for scanned datasheets
- Full eCl@ss/ETIM taxonomy trees + multi-language descriptions
- Image enrichment: product-photo attribute extraction
- Incremental catalog sync + enterprise SSO for review workflow

Provide links to your:

1. GitHub Public Repository
2. Demo Video Link (3 Minutes)
3. Working Prototype Link

Demo Video:

https://github.com/Rahulxoxo69/Statiz/blob/main/demo/demo_video.mp4

Working Prototype:

<https://day-atm-become-bureau.trycloudflare.com>

(live review console — search, filter, approve, export CSV; substitutes API at /api/substitutes/BRG-BEA0000)